

MUNIB AHMAD

Data Scientist & AI / LLM Engineer | Generative AI, RAG & Multi-Agent Systems

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Data Scientist and AI Engineer specializing in end-to-end machine learning and generative AI systems, from data pipelines and predictive modeling to production-grade LLM applications. Experienced in building retrieval-augmented generation (RAG) pipelines, fine-tuning transformer models, and designing multi-agent architectures using LangChain, LangGraph, AutoGen, and CrewAI. Skilled in taking models from research and experimentation to deployed, user-facing applications with Flask, FastAPI, and Streamlit. Strong foundation in statistical modeling, deep learning, and explainable AI, backed by IBM and DeepLearning.AI professional certifications and hands-on internship and project experience.

TECHNICAL SKILLS

Programming & Core CS: Python, SQL, C++, JavaScript, HTML/CSS, Object-Oriented Programming, Data Structures & Algorithms

Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Ensemble Methods (Random Forest, XGBoost, CatBoost), Model Evaluation & Hyperparameter Tuning, Cost-Sensitive Threshold Optimization, Explainable AI (SHAP, LIME)

Deep Learning: CNNs, RNNs/LSTM, ANNs, Transfer Learning, Time Series Forecasting, Neural Network Optimization, TensorFlow, Keras, PyTorch, Object Detection and Tracking, Semantic Segmentation

Generative AI & LLM Engineering: Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Fine-Tuning (LoRA/PEFT), Hugging Face Transformers, Tokenization, Model Pretraining

Agentic AI & Orchestration Frameworks: LangChain, LangGraph, LlamaIndex, AutoGen, CrewAI, BeeAI, Multi-Agent System Design, Tool-Calling, Model Context Protocol (MCP), AI Agents

Application & API Development: Flask, FastAPI, Pydantic, Streamlit, Gradio, RESTful API Design, Web Scraping (BeautifulSoup)

Databases & Vector Stores: MySQL, SQLite, Chroma, FAISS, Semantic Search & Embedding Retrieval

Data Analysis & Visualization: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Plotly, Power BI, Exploratory Data Analysis, Hypothesis Testing, Statistical Inference

Tools & Platforms: Git/GitHub, VS Code, Jupyter Notebook, Google Colab, Anaconda Navigator, PyCharm, IBM Watson Studio, MS Word

Professional Skills: Project Management, Cross-Functional Communication, Critical Thinking, Multitasking, Leadership, Time Management

PROFESSIONAL EXPERIENCE

Data Scientist — Arch Technologies

Remote Internship

- Analyzed multi-domain business datasets covering customer behavior, product preferences, survival outcomes, and time-indexed usage patterns to surface actionable insights for stakeholders.
- Built and validated unsupervised segmentation models (K-Means, hierarchical clustering) to group customers by behavioral and transactional similarity, informing targeted engagement strategies.
- Developed supervised predictive models to estimate survival/retention probabilities and forecast time series trends, iterating on feature engineering and model selection to improve accuracy.
- Used Python, Pandas, NumPy, and Scikit-learn throughout the pipeline — from data cleaning and exploratory analysis to model training, cross-validation, and result interpretation for non-technical stakeholders.

Data Science & Analytics Intern — DevelopersHub Corporation

Remote Internship

- Delivered a full spectrum of ML workflows — classification, clustering, regression, and time series forecasting — across multiple business-focused case studies.
- Applied explainable AI techniques (**SHAP, LIME**) to interpret model predictions and communicate feature-level drivers of outcomes to stakeholders.
- Implemented cost-based decision threshold optimization to align classification models with real-world financial trade-offs rather than default accuracy metrics.
- Prototyped and deployed interactive model-facing dashboards in Streamlit, enabling non-technical users to explore predictions and evaluation metrics directly.

AI / LLM ENGINEERING & AGENTIC AI PROJECTS

- UOS Hostel Assistant — RAG Chatbot** (*Python, LangChain, Flask, Streamlit, Chroma, Mistral*) — Built a retrieval-augmented chatbot that answers University of Sargodha hostel queries strictly from the official handbook using a LangChain Parent-Document Retriever, multi-turn conversational memory, and inline source citations to prevent hallucination; shipped as both Flask and Streamlit applications.

- **VisionGuard AI — YOLOv8 × Mistral Fusion** (*YOLOv8, ByteTrack, Flask, Streamlit, Mistral LLM*) — Designed a real-time object detection and multi-object tracking pipeline that feeds structured event data into a Mistral LLM to auto-generate exportable PDF incident reports; deployed as a CLI tool, Flask web app, and Streamlit app for flexible use cases.
- **GPT-2 From Scratch — LLM Pretraining & Fine-Tuning** (*PyTorch, Hugging Face Transformers*) — Implemented and pretrained a GPT-2 architecture from scratch via self-supervised next-token prediction on the AG News corpus, then fine-tuned the model for email spam classification, achieving 93% test accuracy.
- **Dialogue Summarization Studio — LoRA vs. Adapters vs. Prompting** (*FLAN-T5, LoRA, PEFT, Streamlit*) — Benchmarked zero-shot, one-shot, and few-shot prompting against LoRA and Adapter-based fine-tuning of FLAN-T5-Large for dialogue summarization, then deployed the top-performing configuration behind a Streamlit interface.
- **DeepScan — Deepfake Detection Web App** (*Flask, EfficientNetB0, BiLSTM, InsightFace*) — Engineered a deepfake video detector combining EfficientNetB0 spatial features with a BiLSTM temporal model over InsightFace SCRFD face detection and 106-point landmark biometrics, returning a confidence-scored verdict via a Flask interface.
- **Smart Attendance System — Face Recognition AI** (*DeepFace (FaceNet), OpenCV, Streamlit*) — Built a real-time facial-recognition attendance system for University of Sargodha's Iqbal Hall night attendance using FaceNet embeddings and OpenCV, with automated CSV logging and PDF report generation through a Streamlit interface.

DATA SCIENCE PROJECTS

- **Loan Default Prediction with Business Cost Optimization** (*Logistic Regression, Random Forest, CatBoost*) — Built a cost-sensitive loan default classifier on 32,500+ applications, tuning the decision threshold against asymmetric false-positive/false-negative costs to minimize expected financial loss rather than optimizing raw accuracy.
- **Time Series Forecasting — Stock Market Data** (*ARIMA, SARIMA, Linear Regression, yfinance, Streamlit*) — Compared classical statistical forecasting models against feature-engineered regression to predict stock closing prices from live market data, evaluating stationarity, seasonality, and forecast error across approaches.
- **Potato Disease Classifier** (*TensorFlow, MobileNet, CNN, Streamlit*) — Trained a MobileNet-based convolutional neural network to classify potato leaf images into Early Blight, Late Blight, and Healthy categories, then deployed the model for real-time inference on new images.
- **Laptop Price Predictor** (*Scikit-learn, Random Forest, Extra Trees, Decision Trees, Streamlit*) — Developed a Voting Regressor ensemble to predict laptop prices from specifications (brand, RAM, storage, CPU, GPU, display), combining multiple base learners to improve prediction stability over any single model.
- **Customer Segmentation on Real-World Retail Data** (*Python, RFM Analysis, K-Means Clustering*) — Segmented customers in an online retail dataset using Recency-Frequency-Monetary (RFM) analysis combined with clustering, identifying distinct purchasing-behavior groups to support targeted marketing strategy.
- **Term Deposit Subscription Prediction** (*Logistic Regression, Random Forest, EDA, Streamlit*) — Predicted client subscription to term deposits from bank marketing campaign data through exploratory data analysis and comparison of multiple classification models to identify the strongest predictors of conversion.

EDUCATION

BS Computer Science — University of Sargodha CGPA 3.84 / 4.0, 7th Semester	2023 – 2027
Intermediate (ICS) — Superior College, Quaidabad 940 / 1100	2021 – 2023

CERTIFICATIONS

- IBM RAG and Agentic AI Professional Certificate — IBM / Coursera
- IBM Generative AI Engineering with LLMs Specialization — IBM / Coursera
- Deep Learning Specialization — DeepLearning.AI / Coursera
- IBM Data Science Professional Certificate — IBM / Coursera
- Data Science for Professionals — Coursera

ACHIEVEMENTS

- Honhaar Scholarship (CM Undergraduate Scholarship)
- Position Holder Scholarship, University of Sargodha
- PM Laptop Scheme Recipient (2025)
- Tehsil-Level Position Holder, Intermediate